

Homework #3

Data Intelligence Project

Instructions

Prompt

In your Functional Brief (HW#1) you found where AI could help, and in the Document Intelligence Project (HW#2) you put your function's documents to work. Now you turn to its numbers. You will apply the data-analysis techniques from AI.135, cleaning, validation, exploration, and analysis to a dataset from your function and carry it all the way to a recommendation a busy leader can act on. This builds on the Week 5 readings: Andrej Karpathy's [How I Use LLMs](#) (the data-analysis and tool-use sections) and MIT Sloan's [Practical AI implementation](#).

- **Anchor to your function.** Use the dataset for your function and company and tie it to one of the AI opportunities you identified in your Functional Brief, which is ideally the same workflow you analyzed in HW#2, so your documents and data tell one story. Before analyzing, confirm what every field actually means.
- **Treat cleaning as foundational.** MIT Sloan's success stories stress "foundational investments" like data cleansing before AI projects can deliver value. Use AI alongside Google Sheets to clean and validate the data, and note the issues you found and fixed. Skipping this is how good analysis goes wrong quietly.
- **Make the AI do the math — don't trust a number it just types out.** With documents, the danger was a claim that sounds right; with data, it is a **confident wrong number**. So have the AI work from your actual dataset — use ChatGPT's data-analysis feature, or upload your file and ask it to analyze it — so the charts and numbers come from your data instead of a guess. Then spot-check a few of the results against the spreadsheet yourself before you trust them.
- **Build a choice set, then recommend.** Strong analysts don't hand a leader a single answer. Drawing on MIT Sloan's **intelligent choice architectures**, present two or three options with their trade-offs, then recommend one and explain why. Tie your recommendation back to the function goal from your Functional Brief.
- **Produce the one-page executive recommendation.** Distill everything onto a single page that a busy leader could act on without redoing your work: what the data shows, what it means, the options, and your recommendation with visuals that support the point.
- **Document your process and governance.** Link to your working Google Sheet or notebook so a colleague could retrace your steps, plus a short note on the tools and model tiers you used and how you verified the numbers. Add one governance consideration: who should act on this recommendation, and where does a human stay in the loop? This continues the guardrails-and-governance thread from HW#1 and HW#2.

With documents, the risk was a plausible claim; with data, it is a confident wrong number. Make the AI show its work and verify any figure you would put in front of a leader.

Submission Instructions

- Submit your one-page executive recommendation as a Google Doc on Campuswire.
- Include a shareable link to your supporting analysis (Google Sheet or notebook).
- Label your method note “Tools, Method & Verification,” so it is easy to find, and include your one decision-rights note there.
- Late submissions follow the course Late Assignment Policy.

Grading Rubric

100 points total

#	Category & Target Behavior	Excellent	Good	Poor	Missing
1	Dataset Selection & Relevance (12 pts) <i>Uses a dataset tied to the function and an opportunity from HW#1, and understands every field.</i>	Well-chosen dataset clearly linked to a real decision; fields understood.	Reasonable dataset; link or field understanding partial.	Dataset thin, off-topic, or poorly understood.	No dataset identified.
2	Data Cleaning & Validation (18 pts) <i>Cleans and validates as a foundational step; documents issues found and fixed.</i>	Thorough cleaning with issues identified and resolved transparently.	Reasonable cleaning; documentation light.	Little cleaning; data problems ignored.	No cleaning attempted.
3	Analysis & Verification (22 pts) <i>Has the AI work from the actual data rather than guessing numbers; surfaces trends and anomalies and spot-checks results.</i>	Analysis clearly comes from the data and is spot-checked; findings sound.	Mostly works from the data; checking uneven.	Relies on numbers the AI just stated; little checking.	No real analysis.
4	Choice Set & Recommendation (25 pts) <i>Presents options with trade-offs, then a clear, bias-aware recommendation aligned to the function goal.</i>	Clear option set with trade-offs and a well-justified recommendation tied to function goals.	Recommendation present; options or trade-offs thin.	Single unsupported answer; no trade-offs.	No recommendation.
5	Executive Communication (13 pts) <i>Delivers a clean one-page brief with effective visuals.</i>	Crisp, genuinely one page; visuals aid the point.	Mostly clear; minor clutter.	Hard to follow or far over length.	No brief.
6	Reproducibility & Governance Notes (10 pts) <i>Working link plus a note on tools/model tier,</i>	Fully reproducible with shared link and a thoughtful governance note.	Most elements present; governance note thin.	No working link or method note.	No documentation.

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	<i>verification, and one decision-rights consideration.</i>				